1) Find 10 different model providers and their models released in last 12 months

Details to collect:

Model Provider | Model | Release Date | Cut off Date | Cost of tokens | Context Window | Type | When to use this model |

Provider	Model	Release Date	Cutoff Date	Cost of Tokens*	Context Window	Type	When to Use
OpenAI	GPT-4.5	Feb-25	2024	Premium tier, higher than GPT-4o	128K	Frontier dense	Research, reasoning-heavy tasks
OpenAI	o3 (Reasoning)	Apr-25	2024	Similar to GPT-4o pricing	200K	Reasoning model	Complex chain-of-thought, tool use
Anthropic	Claude 3.5 Sonnet	Oct-24	2024	Lower than Opus	200K	Frontier balanced	General-purpose, cost-efficient
Anthropic	Claude 4 Sonnet	May-25	2024	Premium	200K	Frontier	Advanced reasoning, coding
Google	Gemini 2.5 Pro	Mar-25	2024	API pricing, premium	1M	Frontier multimodal	Long-context reasoning, multimodal tasks
Google	Gemma 2 (9B/27B)	Jun-24	2024	Free (open weights)	8K	Open-source	Lightweight research, fine-tuning
Meta	Llama 3.1 (405B)	Jul-24	2024	Free (open weights)	128K	Open-source frontier	Custom deployments, tool use
Mistral	Codestral 25.01	Jan-25	2024	$0.10 per 1M tokens	256K	Open-source coding	Code generation, large-context coding
Microsoft	Phi-4	Dec-24	2024	Free (open weights)	16K	Small model	STEM reasoning, efficient tasks
Alibaba	Qwen 3 (0.6B–235B)	Apr-25	2024	Free (open weights)	128K	Open-source	Multilingual, math/code tasks
xAI	Grok-2	Nov-24	2024	API pricing	128K	Frontier	Real-time info via X/Twitter
DeepSeek	DeepSeek V3	Dec-24	2024	Open, $5.5M training cost	128K	Open-source frontier	Enterprise-scale reasoning
Databricks	DBRX	Mar-24	2024	Free (open weights)	32K	Open-source MoE	Enterprise AI, scalable workloads

2) Read 2–3 articles about how transformer works?

Goal to understand:

(a) Self-attention mechanism

(b) Vector embedding

(c) Feed forward Nueral Network

1. "Attention Is All You Need" (Vaswani et al., 2017, original paper explained in blogs/tutorials)

- **Self-Attention:** Each word in a sentence looks at every other word, assigning weights that show how much one word should influence another. This allows the model to capture long-range dependencies without recurrence.
- **Vector Embeddings:** Words are represented as dense vectors in high-dimensional space. These embeddings encode semantic meaning, so "king" and "queen" are close in vector space.
- **Feed-Forward Network:** After attention, each position passes through identical fully connected layers. This adds non-linear transformations, helping the model learn complex patterns.
- **Takeaway:** Self-attention replaces recurrence, embeddings give meaning, and feed-forward layers refine the representation.

2. "Illustrated Transformer" (Jay Alammar, 2018, updated tutorials)

- **Self-Attention:** Visualized as queries, keys, and values. Each word generates a query vector, compares with keys of other words, and produces weighted values.
- **Embeddings:** Input words are first converted into embeddings, then positional encodings are added so the model knows word order.
- **Feed-Forward:** Each token's representation is independently transformed by a small neural network, adding depth and expressiveness.
- **Takeaway:** The diagrams make clear how attention distributes focus across words, embeddings anchor meaning, and feed-forward layers enrich features.

3. "Transformers Explained" (Stanford CS Blog, 2024)

- **Self-Attention:** Enables parallel processing of sequences, unlike RNNs. It scales well and captures contextual meaning efficiently.

- **Embeddings:** Serve as the bridge between raw text and numerical computation. They allow semantic similarity to be computed algebraically.
- **Feed-Forward:** Acts like a universal function approximator at each position, ensuring the model can capture non-linear relationships.
- **Takeaway:** The combination of attention + embeddings + feed-forward layers makes transformers powerful for language, vision, and multimodal tasks.

✨ **Key Insights**

- **Self-Attention = Context awareness** (words influence each other dynamically).
- **Embeddings = Semantic meaning** (numerical representation of words).
- **Feed-Forward = Non-linear refinement** (adds expressive power at each position).

3) Connect with your group learners and connect with their LinkedIn profiles → done